

Neonatal and Preterm Infant Growth Assessment



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KEYWORDS

• Preterm • Newborn • Infant • growth • Anthropometry • SGA • LGA • Nutrition
• z-scores • length • head circumference • weight

KEY POINTS

- Historically, the comparison of infant growth between sites has been difficult because consistent growth metrics have not been established for universal use.
- Infant growth assessments should include serial measurements for weight, head circumference, and length. These changes need to be evaluated with realistic goals that have been set with consideration of the infants' genetic potential, prenatal influences, nutritional history, and systemic illnesses.
- If an infant does not show weight gain or head or length growth parallel to established growth curves after the postnatal weight loss phase, first ensure that nutrition has been optimized, and evaluate for genetic and prenatal factors, social determinants, and biological disruptive factors, such as brain injury and other systemic illnesses.
- All anthropometric measures show normal biological variation, and therefore, an individual with size at an extreme (outside of the 3rd and 97th percentiles) and/or unusual growth rates for short periods of time could be normal for an individual.

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- Many infants show brief periods of weight loss or slower weight gain during severe illness. If nutritional intake is appropriate per recommended intakes, then the infant should be assessed, as nutrition is appropriate and no changes would be needed in their care. Achievement of birth percentiles, or achievement of any other specific size, are not appropriate goals.
- Clinical and research needs to assess preterm infant growth differ. Growth assessments are used clinically to understand individuals' growth relative to their individual genetic potential and morbidity status. For research purposes, growth of groups needs to be quantified using meaningful metrics.

INTRODUCTION

Infants born at full term typically take about 3 to 4 months to double their weight. In contrast, preterm infants can double their birth weight in 8 weeks with growth rates per kilogram that are more than twice that of infants born at term gestation.^{1,2} Neonatal researchers and practitioners now recognize that to achieve appropriate growth, preterm infants need to receive adequate nutrients intake throughout their neonatal intensive care unit (NICU) stay. Preterm infants commonly accumulate nutrient deficits during routine NICU care, which prevents them from achieving their growth potentials.^{3,4} Growth assessment remains the most important key to assess nutrition adequacy of preterm infants.

Increasingly, there is evidence indicating a strong association between growth and neurodevelopmental outcomes of preterm infants.⁵ Slower growth rates during the NICU stay and after hospital discharge have been associated with adverse outcomes.⁵ Although growth at rates similar to intrauterine rates is necessary to achieve good neurodevelopmental outcomes in preterm infants, the attainment of superior growth cannot overcome adversities related to social determinants of health, maternal complications of pregnancy, and neonatal morbidities (**Table 1**). Preterm infants in NICUs face several challenges, such as neonatal stress, brain injury, necrotizing

Table 1 Contributors to altered infant growth	
Prenatal	Social determinants Maternal health (medical conditions, such as lupus, anemia, clotting problems, hypertension, diabetes, medications, smoking, alcohol, drugs) Infant genetic potential/inherited size Genetic disorders Multiple pregnancy TORCH infection (toxoplasmosis, rubella, cytomegalovirus, human immunodeficiency virus, syphilis) Maternal weight/weight gain
Neonatal/ postnatal	Social determinants of health Infant genetic potential/inherited size Morbidities: brain injury (which includes intraventricular hemorrhage and periventricular leukomalacia), patent ductus arteriosus, bronchopulmonary dysplasia, necrotizing enterocolitis, sepsis Neonatal stress Nutrition (inadequate nutrient intake, limited oral feeding ability)

enterocolitis, sepsis, and bronchopulmonary dysplasia (BPD). All these challenges can affect growth velocity. Therefore, it is critical to ensure that inadequate nutrition and suboptimal growth do not further contribute to the difficulties that preterm infants may face.

WHAT PRETERM INFANT GROWTH SHOULD WE AIM FOR?

Expert groups recommend that preterm infants should grow and accumulate nutrients similar to the best estimates of how the fetus grows and accretes nutrients.^{6,7} Both the American Academy of Pediatrics (AAP) and the European Society for Pediatric Gastroenterology, Hepatology, and Nutrition (ESPGHAN) also recommend that the nutrition goals for the preterm infant are to provide nutrients to achieve satisfactory neurodevelopment.^{6,7}

For growth charts, the AAP endorses both intrauterine (those based on estimated fetal growth) and postnatal growth charts (those that show the actual growth of infants born prematurely). The Intergrowth 21st Project has developed a set of postnatal growth charts based on the patterns of how preterm infants with “an uncomplicated intrauterine life and low neonatal and infant morbidity” grew.⁸ The project included growth data on only 28 infants born less than 34 weeks and 12 infants less than 33 weeks.⁹ Therefore, the AAP recommends that the Intergrowth Charts not be used for infants less than 36 weeks.¹⁰ Preterm infant nutrition support and medical care have been changing over time, and so it is not clear which population in time and which medical and nutrition practices are the best ones to use as a goal to develop postnatal charts. Once adequate sample size postnatal growth charts are established, they will provide a useful adjunct for preterm infant growth monitoring so that health care providers can use both types of charts to compare each infant’s growth patterns to both fetal and preterm infant references.

Low growth rates of preterm infants and small attained sizes after term age for weight, length, and head circumference have been repeatedly noted to be associated with adverse neurodevelopmental outcomes.⁵ Experts do not recommend the achievement of any specific percentiles for weight, length, or head growth before or by 40 weeks, and they do not recommend return to birth weight percentile after the postnatal weight loss.^{6,7} Preterm infants are not likely to gain lean mass more rapidly than fetal growth rates. Gaining weight to return to birth percentiles, which would require gaining at greater than estimated fetal rates, is often a gain of additional body fat only.

PRETERM INFANT GROWTH ASSESSMENT REQUIREMENTS

Adequate infant growth assessments require serial growth measurements, consideration of prebirth influences, respect of an infant’s genetic potential, and consideration of influences of neonatal morbidities (see [Table 1](#)). For ideal growth monitoring, it is important to have realistic goals, try to understand the infant’s genetic potential, evaluate nutritional needs, and ensure compliance with evidence-informed guidelines regarding daily nutrient intakes. Some infants may need additional investigations to understand causes of deviant growth patterns.

Health care providers need to focus on overall pattern of head and length growth and weight gain rather than striving to achieve a specific percentile by any specific age. There are normal biological variations in size for all anthropometric measures; therefore, an individual with size at an extreme could be normal for an individual.³⁵³⁸ Growth patterns for all anthropometric parameters (ie, weight, length, and head circumference) should be considered individually and in combination. Without serial

measurements and an understanding of an infant's growth patterns and preterm infants' expected growth patterns, it is possible to make incorrect growth assessments. If an infant is assessed to be small or large but is already making appropriate catch-up or catch-down growth toward his or her genetic potential and nutrient intake meets the recommended intakes, then changes for his or her nutrition care might not be warranted. If an infants' growth pattern is as expected given the postnatal weight loss and their morbidities, and their nutrition is equal to recommended intakes, then the infant should be assessed as growing appropriately and no changes would be needed in their care. In contrast, if an infant is found to not be achieving recommended nutrition intakes, then regardless of their growth pattern, their nutritional care may need optimization. Some situations are challenging to assess individual infants' nutrition needs and intakes. Examples of challenges include estimating energy needs of infants with BPD and estimating human milk protein and energy content.

GROWTH ASSESSMENT IN LIGHT OF NEONATAL MORBIDITIES

Many neonatal morbidities have important influences on preterm infant growth. Franz and colleagues¹¹ found that growth to NICU discharge and development at 5 years of age were both associated with severe brain injury, duration of mechanical ventilation, retinopathy of prematurity, and the education of the mother. Of importance, rather than adjusting away the effects of these adversities the infants experienced, they reported the magnitude of different morbidities on neurodevelopmental outcomes to show their relative associations. They found that "the combined contribution of severe [brain injury] and of prolonged mechanical ventilation [on neurodevelopmental outcomes] was >9 times greater than the combined contribution of growth."¹¹ Others have also found brain injury to be among the most important predictors of adverse neurodevelopment.¹² Of note, Franz and colleagues¹¹ found head growth, both during NICU stay and after discharge, to be associated with neurodevelopment. Researchers in New Zealand found that early preterm infant protein intakes predicted head growth and neurodevelopmental outcomes.¹³

The actual causal pathway for poor neurodevelopment in units where good nutritional care is provided is likely primarily due to adverse events, including severe brain injury and prolonged mechanical ventilation, which cause both poor growth and poor development.^{11,1438} In units with good nutrition care, growth is more likely a marker for neonatal adversities than a simple reflection of poor nutrition.

PRETERM GROWTH PATTERNS: WHAT IS USUAL GROWTH AND INAPPROPRIATE GROWTH AND WHAT NEEDS TO BE DONE ABOUT IT?

To be able to assess preterm infant growth, it is important to have a good understanding of the usual patterns. Most, but not all, preterm infants lose weight in the first days of life due primarily to water loss as they transition from a water to an air environment (Fig. 1A–D). Many preterm infants do not grow in length or head circumference in the first week of life, so all 3 growth parameters often cross percentiles downwards on growth charts. If adequate nutrition is established, tolerated, and maintained, and when preterm infants avoid severe neonatal morbidities, they then tend to grow approximately parallel to growth chart curves, which shows growth tracking similar to estimates of fetal growth, as shown in growth charts, such as the Fenton preterm growth charts.¹⁵ Prenatal factors, genetic potential, establishment of adequate nutrition, feeding tolerance, and neonatal morbidities will then play a role in the patterns of growth (see Table 1). The following examples of challenging preterm infant growth assessments may help understand the variables that need to be considered.

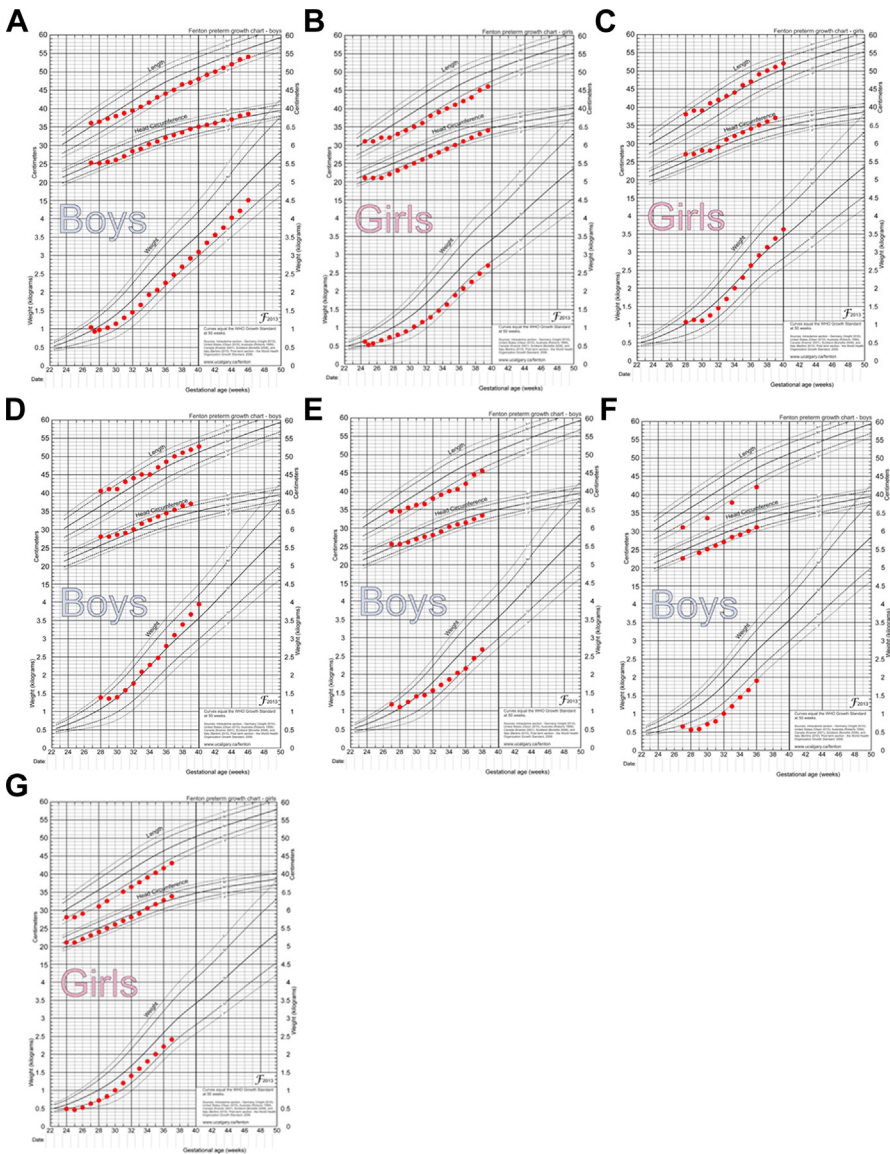


Fig. 1. Seven patterns of appropriate growth. (A) AGA baby with normal growth pattern. (B) AGA infant with "misnomer extrauterine growth restriction."³⁵ (C) AGA infant with "upward crossing" due to genetic potential after in utero growth restriction. (D) LGA infant with good growth due to genetic potential. (E) LGA infant returning to normal growth, maternal diabetes. (F) SGA infant with growth parallel growth. (G) SGA infant regained birth weight percentile. (Plotted using Canadian Pediatric Endocrinology Group. Growth chart plotter for preterm infants 22 to 49 weeks based on Fenton 2013. https://cpeg-gcep.shinyapps.io/prem2013_DDE/ (last accessed Nov 10, 2021).)

ASSESSING SIZE AT BIRTH USING GROWTH CHARTS

There are 2 key purposes for which health care providers use growth charts for infants: (i) to assign size for gestational age as small (SGA), appropriate (AGA), or large (LGA),

and (ii) to monitor growth. Perhaps the most immediate use for identifying infants by their SGA and LGA status is to identify infants at birth who need additional monitoring because SGA and LGA infants are at higher risk of hypoglycemia. Several tools are currently available to assess preterm infant's birth size and postnatal growth, such as Olson and Fenton growth charts. Olsen charts are made using American intrauterine size data up to 41 weeks' gestation,¹⁶ whereas Fenton intrauterine size data are based on a 6-country meta-analysis of almost 4 million infants up to 42 weeks' gestation.¹⁵ For assigning size for gestational age for term infants, the World Health Organization (WHO) growth standard² at birth could be used because infants born between 37 and 42 weeks were included in the WHO study.

THE VALUE OF ASSESSING Z-SCORES IN PRETERM INFANTS

How useful are the z scores to quantify preterm infant growth? What amount of z-score loss is too much? We do not really know. Z-score losses are variable in any cohort of preterm infants. At this point in time it is not clear how useful changes in z-scores are.

Z-scores define a person's anthropometric measurements (eg, weight, head circumference or length) compared with a growth chart in terms of the number of standard deviations below or above the growth chart median (50th percentile). Changes in z-scores provide information about whether the person's measurement is increasing, decreasing, or staying parallel. Z-scores are considered a good way to quantify a person's anthropometric measurements because they are meaningful for any size, whereas percentiles are not very meaningful at the extremes of size. Numerical values for percentiles at the extremes are small (for example, from -2 to -3 z-scores percentile equivalent values are 0.2 and 0.001, respectively), whereas z-scores remain the size of 1 standard deviation across the distributions.

Neonatal nutrition experts have recently recommended that preterm infant weight losses greater than 2 z-scores is an indicator of severe malnutrition.¹⁷ They also suggest that preterm infant weight losses more more than 1.2 z-scores suggest moderate malnutrition, and losses greater than 0.8 z-scores suggest mild malnutrition. These cut points have not been validated, and there are 2 reasons these cut points need validation: they are arbitrary cut points, and even the healthiest well-nourished preterm infants sometimes lose up to 2 weight z-scores in their first weeks of life.¹⁸

Rochow and colleagues¹⁸ found that the "healthiest" protocol-nourished less than 30-week gestation preterm infants weights decreased by an average of 0.8 z-scores by 3 weeks of age after their postnatal extracellular water weight loss. These healthy protocol-nourished infants had a considerable range and variability in their z-scores losses; some of these infants had a loss of 2 zscores, whereas others had a small gain in z-scores by 3 weeks of age. Although z-scores and changes in z-scores are useful to quantify infants' sizes, losses, and growth, any arbitrarily-defined growth metric will misclassify some individual infants. Determining an appropriate z-score weight loss guideline requires further validation work on z-score changes to determine whether there is a cutoff associated with adverse outcomes. Although changes in z-scores have the advantage that they adjust for sex and gestational age,¹⁹ changes in z-score expectations may need to be different for SGA (Fig. 1F, G) infants and perhaps infants born to mothers who had diabetes (Fig. 1E). Because of the current lack of validation for using z-score changes as predictors of health and neurodevelopmental outcome, individuals should not be labeled to their parents as having poor growth owing to a large changes in z-scores.²⁰

The figures in this article assume good measurement of length as well as head circumference and weight. Of these 3 parameters, length is the most poorly measured. To be able to adequately assess infant growth, their length needs to be measured periodically by 2 people using a length board and good technique.^{6,21}

Appropriate Growth

AGA baby with normal growth pattern

Characterized by postnatal weight loss and slower head and length gain in early weeks, all 3 parameters often shift lower on growth chart curves **Fig. 1A**. Head circumference is first to show recovery toward the median, with final percentiles determined by genetic, nutritional, and medical variables. At week 35, this infant's weight gain velocity slowed to 9 g/kg/d (calculated using the average method)²²; however, weight gain for the week before was 22 g/kg/d, and the overall pattern shows good growth in all 3 parameters at approximately the estimated intrauterine rate. Recommended weight gain velocity to 36 weeks based on fetal growth estimates is 15 to 20 g/kg/d.²² The temporary slower rate of 9 g/kg/d is not a concern, because it was a short-term fluctuation and his growth remained appropriate. Weight and length recovery to genetically determined patterns usually occur after term age (40 weeks' postmenstrual age). This baby's birth percentiles were all approximately at the 60th percentile, which does not suggest any concerns. Head circumference has returned close to the 60th percentile at about 41 weeks, while weight is beginning to show some catch up at this age; the birth percentiles are not necessarily the goals because the baby's growth is no longer directly influenced by the uterine environment and so influences are different. This infant is growing appropriately.

AGA infant with "misnomer extrauterine growth restriction"

This 24-week gestation baby had a typical growth pattern for a preterm infant, a growth pattern of no concerns **Fig. 1B**. Her weight might be categorized as growth failing because it is below the 10th percentile at 36 weeks; however, that age is too young to show much or any catch-up growth after the postnatal weight loss phase. Her growth is approximately good, parallel to growth chart curves after postnatal weight loss, which means she is meeting her growth goals. Her head and length growth suggest that she may be beginning catch-up growth. Being small (weight, length, head) after term (at 3 to 24 months corrected age) for preterm infants has been associated with poorer cognition.^{1,5} Therefore, growth matters, but size at 36 weeks' postmenstrual age is not predictive of neurodevelopment.^{1,5} This infant should be not be considered to be growth failing or growth restricted on the basis of weighing less than 10th percentile at 36 weeks¹ and should be nourished using pre-term estimated nutrition recommendations.

Using a more comprehensive analysis regarding the diagnostic accuracy of the full range of percentile cut points at 36 weeks for cognitive outcomes, researchers found that across the full range of weight, length, and head circumference measures: no cut point was predictive of cognitive outcomes.¹ Infant growth is very important, as numerous studies have found that the AAP and ESPGHAN recommendation that pre-term infants should grow at intrauterine rates is well supported by evidence.⁵

AGA infant with "up-ward crossing" due to genetic potential after in-utero growth restriction

This baby's birth percentiles were higher for length and head circumference than weight, which may suggest some in utero growth restriction **Fig. 1 C**. Her mother had preeclampsia. Both parents were taller than median heights. Health care providers and parents usually do not know an infant's genetic potential, although parental

heights can assist making an estimate.⁶ This baby shows high placement on the curves for all 3 parameters, with some early before-term catch-up in weight. She regained her weight and head percentiles before 40 weeks, likely because of the combination of being tall based on genetics and in utero growth restriction. SGA preterm infants are more likely to increase their weight percentile while in the NICU than AGA infants.⁴⁹ Babies in the AGA category can be growth restricted relative to their genetic potential.

Some neonatal researchers suggested that the expected goal of preterm infant growth is to have the infant return to their birth weight percentile before NICU discharge.²⁴ However, to regain birth weight percentile after the initial weight loss requires weight gain at rates that are faster than the fetus. Some preterm, particularly SGA, infants do more often than AGA infants regain their birth weight percentile as they catch-up to their lean mass growth potentials. Lean mass growth is influenced by genetics and requires less energy to grow than fat mass²⁵; therefore, growth-restricted infants are able to gain weight faster on the same energy owing to their growing a higher proportion of lean mass.

Large-for-gestational-age infant with good growth due to genetic potential

This LGA 28-week preterm infant showed a growth pattern of a proportionate infant (Fig 1 D). He followed the usual preterm infant pattern of losing some weight postnatally and having a lag in head and length growth, but all 3 of his anthropometric measures are in the upper percentiles. He has not returned to his birth size percentiles by 40 weeks. His growth does not suggest any concerns. Populations of infants of any gestational age are expected to have a wide distribution of percentiles for all anthropometric measures (weight, length, and head). Most of these infants are likely to measure between the 3rd and 97th percentiles. The probability of being smaller than the 3rd percentile or greater than the 97th percentile is small, occurring only among 6% of infants. After the postnatal weight loss that most infants achieve, they will place on the lower percentiles. Although having a birth weight greater than the 90th percentile has been associated with higher rates of hypoglycemia, some of those infants are genetically, not pathologically, greater than the 90th percentile at birth, and are not likely at higher risk of these adverse outcomes. However, some genetically small and large infants are likely to be misclassified by the small, large, and appropriate (SGA/LGA/AGA) categorization.

This infant should continue to be nourished based on evidence-informed nutrition recommendations. This child is not overweight despite weighing in the high percentiles, as his length and head are also in the high percentiles.

Even if this infant's length and head were not also in the high percentiles, it is good practice to continue to provide nutrition within recommended intakes because infants need nutrition support without too much concern about inducing overweight because at older ages preterm infants are at slightly lower risk of overweight.^{26–28}

Large-for-gestational-age infant returning to normal growth, maternal diabetes

This LGA 27-week preterm infant had a higher percentile weight compared with his head and length at birth. His mother had poorly controlled diabetes during her pregnancy (Fig. 1E). At 36 weeks, 9 weeks of age, his placement for weight, head, and length on growth chart curves was similar to a baby that did not have as high a birth weight percentile. His health care team thought that his weight had been elevated at birth because of his mother's diabetes. They were satisfied with his growth and thought he was adapting well. The infant was expected to gradually transition to his genetic potential for growth. This infant's birth weight was likely elevated by his

mother's diabetes, and after birth, his weight gain curve was flatter than usual because of his genetic potential. His placement on the growth chart after 34 weeks was similar to the baby in [Fig. 1A](#).

Fetal growth can be influenced by prenatal morbidities. Two well-known examples include gestational hypertension that can impede growth, and poorly controlled gestational diabetes that can lead to excess fetal weight gain. Once the infant is no longer in the in utero environment, these influences no longer influence infant growth.

Small-for-gestational-age infant with parallel growth

This SGA 27-week preterm infant had a growth pattern common among SGA infants [Fig. 1F](#). His parents had short statures (mother: 150 cm, father: 163 cm), and their son will also likely have a short stature. His small size may be genetic and not pathologic or reflecting restricted growth.

Small infants who are constitutionally small and who have not had their growth restricted are likely to grow in a pattern similar to AGA infants, but they may be somewhat lower on growth chart curves. Although having a birth weight less than the 10th percentile has been associated with higher rates of hypoglycemia, necrotizing enterocolitis, and retinopathy, some of those infants are genetically, not pathologically, less than the 10th percentile at birth, and are not likely at higher risk of these adverse outcomes. This infant should continue to be nourished based on evidence-informed nutrition recommendations. Current nutrition recommendations are not different for SGA compared with AGA infants.

Small-for-gestational-age infant regained birthweight percentile

This SGA 27-week preterm infant regained her weight and head percentiles before term age [Fig. 1G](#). Such recovery is seen more frequently in SGA infants than in AGA infants. This infant's regaining of her birth percentile likely reflects her genetic potential supported larger size that she likely would have achieved if her mother was healthy and nutrient delivery via the placenta had been appropriate. Among very preterm infants (<30 weeks) in the PreM Growth Study,¹ the rate of regaining birth percentiles by 36 weeks was greater among SGA infants, presumably because of these infants showing some catch-up growth toward their genetic potential. The rates of regaining birth percentiles by 36 weeks for weight were only 8% of AGA and LGA infants, but 14% among SGA infants.

Regaining head circumference birth percentiles is not a rare occurrence, presumably because of head sparing, that is due to the infant prioritizing head growth over weight gain as well as due in part to the deformational plagiocephaly that many early preterm infants experience, which can affect measurements.^{29,30} By 36 weeks, 23% of AGA/LGA infants, but 31% of SGA infants of very preterm infants (<30 weeks) in the PreM Growth Study regained their birth head circumference percentiles. The cause of small birth weight for any given infant depends on their genetically determined size, maternal morbidities, the social determinants of health, and likely in many situations is a combination of all 3 of these factors.

Inappropriate Growth Patterns

Appropriate-for-gestational-age infant with poor growth

This AGA 28-week preterm infant showed a concerning growth pattern that did not keep up with the growth chart curves ([Fig. 2A](#)). After discharge at term age, his growth deviated further away from the normative data in the growth charts. Numerous prenatal and neonatal conditions contributed to slow his growth both in the hospital and after discharge (see [Table 1](#)). The health care team, including the dietitian and an

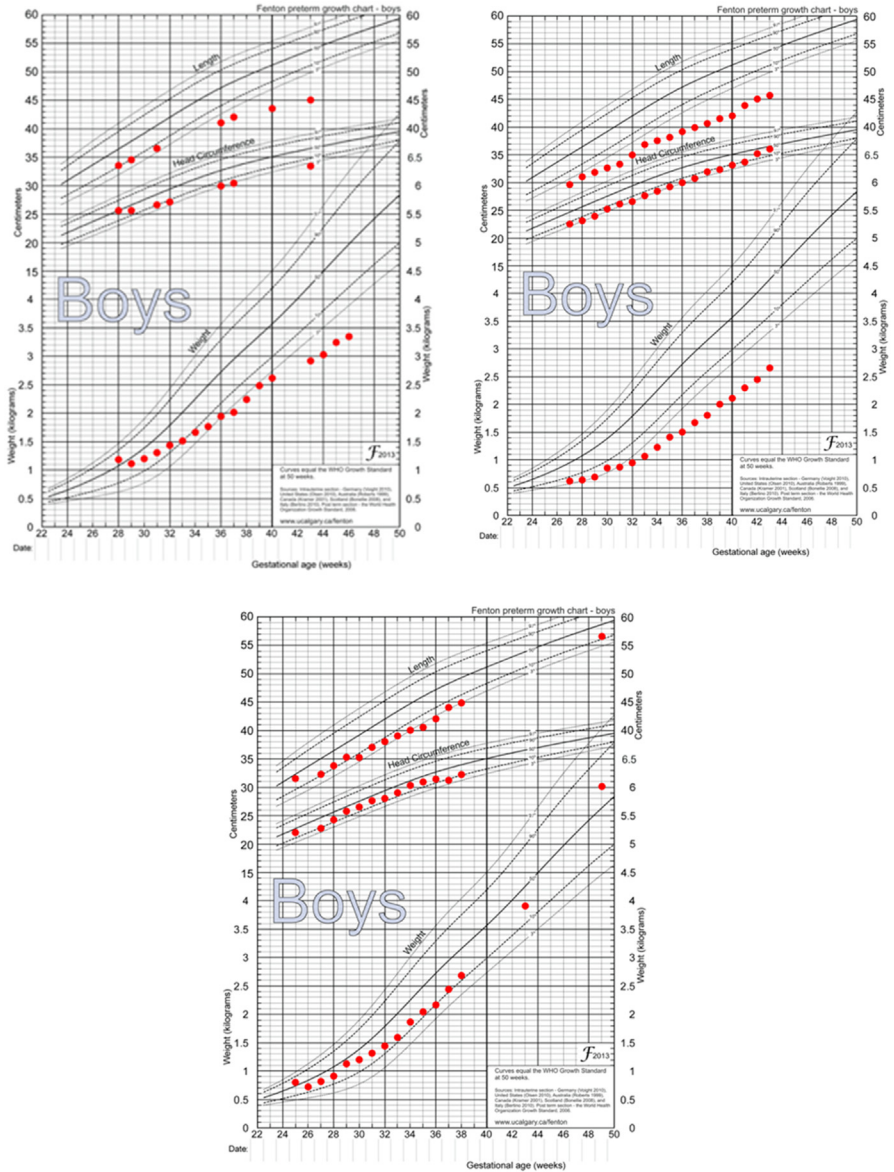


Fig. 2. Three patterns of inappropriate growth. (A) Appropriate-for-gestational-age infant with poor growth. (B) SGA infant who continues to grow poorly: needs investigation. (C) Excessive weight gain (Plotted using Canadian Pediatric Endocrinology Group. Growth chart plotted for preterm infants 22 to 49 weeks based on Fenton 2013. https://cpeg-gcep.shinyapps.io/prem2013_DDE/ (last accessed Nov 10, 2021).)

occupational therapist, needed to ensure that his nutrition was adequate, as suboptimum nutrition could further complicate his growth outcomes.

SGA infant who continues to grow poorly: needs investigation

This infant's prenatal history (maternal or fetal) did not provide any explanation for this infant's growth. His growth did not remain parallel to curves (Fig. 2B). He had some

feeding intolerance and needed time to adapt to enteral nutrition, but fortunately, he had no important morbidities. Although we need to accept, that even with good nutrition, some deviations from the growth curves can happen, this infant continued to deviate away from the growth chart curves for weight and length. This baby needed to be nourished based on evidence-informed nutrition recommendations (because we lack preterm evidence-informed guidance on how to adapt recommendations for situations of poor growth) with continued follow-up to ensure he is able to grow adequately after discharge and a workup to determine whether the infant has an undiagnosed morbidity.

Poor growth in SGA infants needs special attention, particularly in the absence of known causes of poor fetal growth (Fig. 2B). These infants may need further investigations to identify their cause for being SGA. Several rare genetic and endocrine disorders have been reported in preterm infants with severe growth restriction and slow growth velocity. These infants require careful examination of the growth patterns for weight, head, and length, nutrition care received, as well as consideration of additional factors that influence growth, including prenatal contributions (see Table 1). Each poorly growing infant (who does not show rates of fetal growth that fit the normative data) indicates the need for further investigation to determine whether poor growth is modifiable or explainable and to determine whether the infant requires more or different care. The goal of neonatal care is to support young infants to achieve the best they can be.

Excessive weight gain

These data are from an AGA 27-week preterm infant. His mother had poorly controlled diabetes during her pregnancy, and at birth he had a higher percentile weight compared (percentile: 67 z-score: 0.5) with his head (percentile: 29 z-score: -0.6) and length (percentile: 35 z-score: -0.4) Fig. 2C. At 36 weeks, the growth patterns seemed to be normalizing, and he was beginning to show good transition to oral feeding. At 2 months corrected age, the family informed their dietitian that they had been working very hard to feed him as much as they could manage to keep his weight gain high. She found that his weight was now at the 70th percentile (0.5 z-scores), and height at the 15th percentile (-1 z-scores) based on the growth chart.¹⁵ The dietitian reassured the family that their son was doing well and advised that they focus instead on his feeding cues for hunger and satiety to avoid overfeeding. Inconsistent messaging about growth targets from health care providers and from within the caregiver's community can interfere with responsive feeding, overriding the infant's hunger and satiety, and contribute to overfeeding and excessive weight gain (Fig. 2C).

BODY MASS INDEX

Body mass index (BMI; defined as weight in kilograms divided by height in meters squared) has been suggested as a measure to identify appropriate preterm infant growth.³¹ However, caution is needed when BMI is used since rising BMIs for preterm infants could easily be misinterpreted. Two groups of researchers have judged that a clinical application for the use of BMI among preterm infants has not been established.^{32,33} Using BMI has the same limitations as weight assessments because it would not be clear whether increases are due to excess fat or improved lean mass gain; additionally, BMI measurements could increase in preterm infants because of typically slower length gain (see Fig. 1A) than weight gain. A further limitation of using BMIs for any population is that they do not assess body composition but rather simply express weight relative to height. Furthermore, any inaccuracies in length measurement, which are common for infants,²¹ would be amplified. For populations other than preterm infants, higher and increasing BMIs are used as a crude screening tool to identify possible excess body weight. Usually, preterm infant weight catches up

before length so their BMIs would be expected to increase. Rising preterm infant BMIs might be considered by some to indicate too rapid body mass gain when the actual reason for the higher values could be delayed length growth. Wells and colleagues²³ found that preterm infant discharge BMI z-scores were above the reference median because length z-scores were consistently lower than weight z-scores.

BODY COMPOSITION

Higher preterm infant body fat percent observed at term age has been a recent concern among neonatal health care providers; however, evidence suggests that it is temporary and does not persist, so it is not a reason to restrict growth. A recent review summarized body composition data on preterm and term infants between 35 and 65 weeks postmenstrual age (PMA). At 40 weeks, preterm infants had a higher percentage body fat than term infants (16% vs 11%, respectively; $P < 0.001$), but by 52 weeks PMA, percent fat mass was similar for preterm and term infants, averaging 24% to 25%.³⁴ It appears likely that all infants gain body fat over their first postnatal months,^{35,36} and that this occurs at a younger postmenstrual age for preterm infants versus term infants, so it is only different in the short term.

Many are also concerned about lower lean mass for preterm infants at 40 weeks' PMA; however, body composition studies suggest that this is also a temporary phenomenon. The review by Hamatschek and colleagues³⁴ found that preterm infants compared with term newborn infants had significantly lower fat-free mass by 400 g on average at 40 weeks' PMA, but this difference decreased to less than 100 g when both groups were 60 weeks' PMA. It is worth realizing that preterm infants are not likely to gain lean mass beyond fetal growth rates. Therefore, for nonintrauterine growth-restricted infants to regain birth percentiles following their postnatal weight loss and to then continue to grow and reach body composition of a typical term infant would require gaining at greater than estimated fetal rates, which would likely require gaining additional body fat.

A recent study reinforces that NICU nutrition provision is important to achieve good outcomes. Researchers observed that higher energy and protein intakes in the first week of life and protein intakes throughout hospitalization were associated at discharge with higher fat-free mass z-scores but not higher percent body fat z-scores. They also found that higher fat-free mass z-scores at discharge were associated with higher Bayley III scores.³⁷ This finding that preterm infant body fat percentage at discharge was not related to nutrition intakes reinforces that the frequently seen higher percent body fat at term age is likely a postnatal adaption.

NOURISHING INFANTS BASED ON GROWTH PATTERNS AND INFANT FEEDING CUES

Nutrition care support of preterm infants can be divided into 3 phases: parenteral, enteral, and oral feeding. Times of greatest risk of inadequate nutrition include getting established parenteral, enteral nutrition,¹³ and oral feeding as well as the transitions between these phases. Optimizing nutrition during the transition periods is a key to prevent nutrition deficits.

Initially, preterm infants who are parenterally or enterally fed need to be nourished based on the best evidence-informed nutrition guidelines. None of the preterm nutrition guideline groups provide different guidance to nourish infants who are undersized or oversized.^{6,7,38} Whether an infant's growth identifies a concern, nourishing these infants should undergo a transition from infants being provided evidence-informed recommended nutrient and energy amounts in the early weeks, to being supported to feed based on the infant's own cues of hunger and satiety. Many preterm infants will transition from prescribed volume enteral feedings to full oral feeding before

discharge home. Some infants will require feeding strategies to support them to orally feed safely and adequately, and some may require some enteral tube feedings at discharge. Feeding protocols, which include infant-driven progression of oral feeding consider physiologic and neurologic maturity, infant skill, and development^{39–41} and are associated with shorter time to transition to full oral feeding,^{39,40} shorter length of stay,^{40,42} and in one study, improved infant growth.⁴²

Achievement of full oral feedings by 40 weeks PMA was associated with better neurodevelopmental outcomes at 18 to 26 months corrected age¹². Inability to fully feed orally at 40 weeks' PMA may be a simple, clinically useful marker for risk of adverse neurodevelopmental outcomes.¹²

There is a paucity of studies or guiding algorithms to support families through the transition to on-demand feedings around discharge. Transition to cue-based feeding for timing and volumes requires caregivers who can sensitively respond to their infant to provide safe and pleasurable feeding that supports growth and provides a foundation for a positive feeding relationship. There are 2 reasons for allowing children to determine the amounts fed: (i) we never really know a child's individual genetic growth potential and (ii) we need to support them to trust and respond to their own appetite. Parents need support to trust their infant and to overcome the NICU emphasis on volumes consumed and short-term weight gains. The premise/goal of nourishing a child once the child is able should be based on infant cues of hunger and satiety and follow responsive feeding practices, which support the division of responsibility.⁴³ If a child is assessed as too small or too heavy or has growth that deviates from their usual pattern, the division of responsibility guidelines encourages caregivers to respect infant's hunger and satiety. There will be some exceptions for infants who may need supplemental tube feedings as well as those who have rare syndromes whereby hunger-satiety self-regulation is impaired.

RECOMMENDATIONS FOR PRETERM INFANT GROWTH MONITORING

The needs to assess preterm infant growth clinically differ from the needs to summarize growth for research. Clinically, growth assessments are used to understand individuals' growth relative to their individual genetic potential and morbidity status to be able to support the needs of each individual baby. Growth quantification for research purposes is used to quantify growth of groups using meaningful metrics. Historically, neonatology has lacked consistency in the use of growth metrics.^{44,45}

CLINICAL GROWTH MONITORING

1. Aim to meet nutritional needs for all infants to support their growth.
2. Aim for growth approximately parallel to growth curves. Use the growth patterns of all 3 growth parameters (weight, length, and head circumference) plotted on growth charts to assess growth.⁴⁵ "Increasing weight out of proportion to length does not confer developmental benefits" or improve outcomes.^{46,47}
3. Do not be concerned about preterm infant body fat proportion at 40 weeks, as it increases only temporarily as a postnatal event in preterm as well as term infants.^{26,27,34}
4. Accept variability; remember biological variation: it is normal to see a range of sizes and growth rates. Do not aim to exceed any specific percentile or z-score because any infant's genetic potential is unknown and sizes within and, rarely, even outside of the 3rd to 97th percentiles are normal and expected in a population.
5. Do not aim for a return to the birth weight percentile after the extracellular fluid contraction weight loss.

6. Infants should not be considered growth restricted or failing just because their weight is less than 10th or 3rd percentile between 36 and 40 weeks' PMA.¹
7. If nutrition is optimized and growth does not equal intrauterine rates, look for other potential causes (see [Table 1](#)).^{14,20,38}
8. Do not label children to parents/caregivers as growth restricted, malnourished, stunted, wasted, underweight, overweight, or obese, as labels are not helpful for individuals.²⁰ It is better when talking with parents/caregivers to instead use less judgmental terms, when necessary, such as "weight is ahead of length" or "length is not as tall as expected."
9. When a child is able to self-feed, respect the child's appetite and satiety⁴³ and feeding abilities.

RESEARCH GROWTH MONITORING

The purpose of research growth monitoring is to summarize growth of groups of infants for research or audit purposes.

1. Report birth weight, length, and head circumference and z-scores for weight, length, and head circumference. Report the reference data set used to derive the z-scores.⁴⁵
2. Growth velocity calculations (using grams per kilogram per day or changes in z-scores) to compare to intrauterine rates should begin at the nadir or day 7 (because the extracellular fluid contraction/postnatal weight loss phase is not growth, so it should be excluded). Growth velocity can be quantified using grams per kilogram per day up to 36 weeks (use the average or exponential method²²); at ages older than 36 weeks, use grams per day²²
3. For research summaries, z-score changes can be used at any age and for most time periods (after the postnatal weight loss phase), although short time periods are less likely to represent longer-term growth⁴⁸ (see [Fig. 1A](#)). For research, report z-score changes and/or growth velocity to 28 days and/or 36 weeks PMA or discharge, as relevant.⁴⁵
4. Growth velocity from birth can also be useful to document in research, but when begun at birth, growth velocity should not be expected to equal intrauterine rates because of the postnatal weight loss phase.
5. Infants should not be considered growth restricted or failing just because their weight is less than 10th or 3rd percentile between 36 and 40 weeks PMA.¹
6. Ideally, assess neurodevelopment at 2 years or older corrected age. Consider the numerous possible contributors to suboptimal development.^{14,38}

Best Practices

- Without serial measurements and an understanding of preterm infants' expected growth patterns, it is possible to make incorrect growth assessments.
- If an infant is assessed to be small or large but is already making appropriate catch-up or catch-down growth toward his or her genetic potential, and nutrient intake meets the recommended intakes, then changes for his or her nutrition care might not be warranted.
- If an infants' growth pattern is as expected given their postnatal weight loss and morbidities, and their nutrition is equal to recommended intakes, then the infant should be assessed as growing appropriately and no changes would be needed in their care.
- If an infant's growth is slower than growth chart curves, further investigations are needed to explore for nutrition, morbidity, intrauterine or genetic reasons with the actions required depending on the cause.

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